

AUDIO COMPRESSION

Using Wavelets In MATLAB

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Audio frequencies range from 20Hz to 20kHz but these frequencies are not heard in the same way. Frequencies below 20Hz and above 20kHz are very difficult to hear, while those not much more than 20Hz, or not much less than 20kHz, cannot be heard by most people. We often need to process these audio signals for various applications. MATLAB is one of the best signal analysis and signal processing tools.

Audio compression is a very good example of speech and signal processing. We use the Internet for various purposes including entertainment. Audio is common in all entertainment applications. If an audio file size is large, it takes more space to store. Audio/video compression frees up space substantially, which can then be utilised for other purposes. This article describes some important audio compression techniques.

An audio signal sample is taken and analysed using MATLAB for frequency and amplitude. Haar and Daubenches algorithms are applied on the speech signal and the audio is compressed. Audio sizes before and after compression are compared.

The following parameters are compared by the program: Peak signal-to-noise ratio (PSNR), normalised root-mean-square error (NRMSE) and compression ratios. The related mathematical equations are included in this month's EFY DVD for reference.

Haar wavelet algorithm performs the following functions:

1. Selects audio and finds actual signal size
2. Finds amplitude and frequency
3. Creates frame
4. Decomposes the signal spectrum into wavelet
5. Creates psychoacoustic model
6. Inspects the spectrum and finds tones maskers

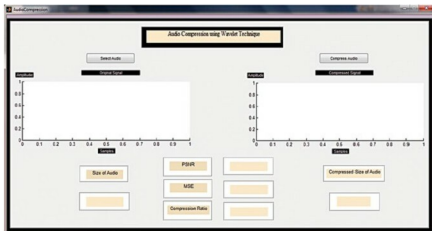


Fig. 1: Program output (Haar wavelet)

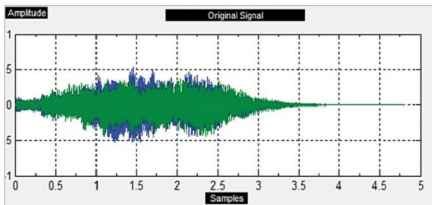


Fig. 2: Original audio signal (size: 414.691kB)

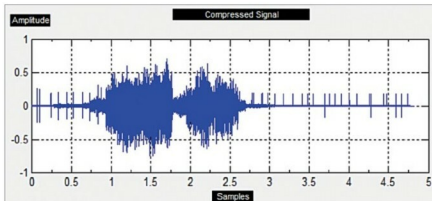


Fig. 3: Haar-wavelet-decomposed audio signal (size: 207.367kB)